

ConstructionSceneReady: Evidence-grounded compilation and task-scene validation for BIM-driven robotic assembly

Zhu Aiyu^{a,*}

^a*Yango University, Quanzhou, 362014, China*

Abstract

Robot-ready construction simulation requires more than converting building geometry into a simulation format. This study asks which building information actually supports a robotic construction task in simulation, and how its absence, error, or imprecision changes task outcomes. We propose ConstructionSceneReady, which compiles IFC source into task-activated multi-scale OpenUSD scenes and validates the composed scene against an explicit construction task-scene contract with evidence-bounded repair. On a frozen 105-case fault-injection partition, the contract recovers all 141 expected findings without false positives; whitelisted repair restores 97.1% of scenes to pristine equivalence and escalates the rest. Constraining a language-model agent to the same whitelist preserves this reliability (96.8% valid, zero incorrect repairs in 525 episodes), while an unconstrained agent produces validator-passing but incorrect scenes in 22.7%. Paired humanoid insertion episodes then measure the task consequence of each information family: symbolic and referential defects are discretely fatal, interface clearance is the most sensitive continuous property, and absolute member-pose errors are absorbed by closed-loop perception. Crossing these outcomes with the contract shows perfect specificity but limited sensitivity to sub-tolerance geometric fidelity; we convert that blind spot into two contract rules verified on an independent test set. Evidence comes from parameterized fixtures and one fixed-base executor with a kinematic grasp scaffold; the conclusions are structural—

*Corresponding author

Email address: zhu.aiyu@yango.edu.cn (Zhu Aiyu)

which information families are fatal, graded, or self-correcting—rather than absolute performance claims.

Acknowledgements

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.